



# Liquid fuels demand and supply model

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Manish hand-back

7 October | Baseline, fuel input tables and decision log

7 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Durban | vopak.com



# On customs trade, 2024 petrol and diesel sales exceed net imports by 7.7 bn litres

Corrected balance | calendar years, billion litres | imports and exports from SARS customs

| Year | Petrol |         |         |                        | Diesel |         |         |                        |
|------|--------|---------|---------|------------------------|--------|---------|---------|------------------------|
|      | Sales  | Imports | Exports | Sales less net imports | Sales  | Imports | Exports | Sales less net imports |
| 2019 | 10.77  | 1.48    | 1.17    | 10.47                  | 12.91  | 5.59    | 1.74    | 9.06                   |
| 2020 | 8.76   | 1.72    | 1.15    | 8.20                   | 11.69  | 6.81    | 0.91    | 5.79                   |
| 2021 | 9.30   | 4.01    | 1.05    | 6.35                   | 12.95  | 9.76    | 0.89    | 4.08                   |
| 2022 | 9.18   | 5.46    | 0.85    | 4.58                   | 12.72  | 11.95   | 0.89    | 1.65                   |
| 2023 | 9.04   | 4.48    | 0.99    | 5.55                   | 12.91  | 12.87   | 0.89    | 0.93                   |
| 2024 | 9.03   | 4.00    | 0.92    | 5.95                   | 11.73  | 10.79   | 0.80    | 1.74                   |
| 2025 | —      | 4.45    | 0.80    | —                      | —      | 12.25   | 0.75    | —                      |

Sales less net imports is what production, stock changes and gaps in sales coverage must together supply. It is not a measurement of production.

Sales: department to 2023; 2024 is FIASA's figure and is unverified; none yet for 2025. Rebuilt by one command from registered inputs; a test checks every year against the customs extract.

## Sources agree on most years; five differences need a recorded choice

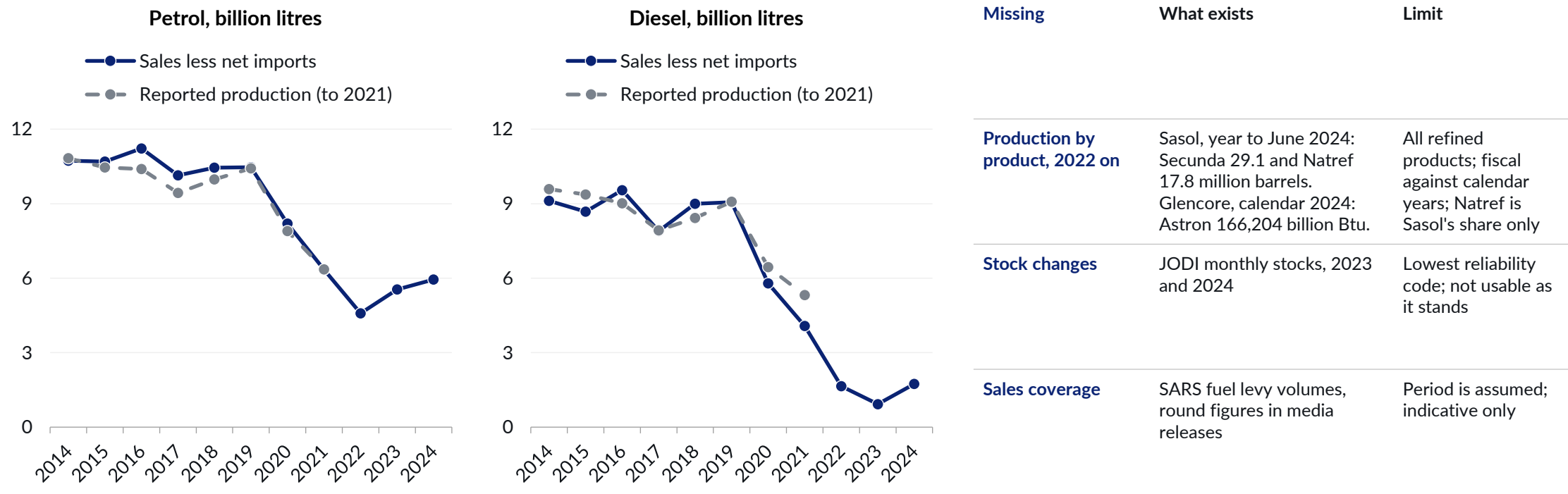
Source differences | billion litres | every source is kept in its own column of the balance file

| Item                                 | What the sources say                                                                                                                                                                                                                                                                              | Used in the balance                                  | Who decides                        |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------|
| <b>2024 sales</b>                    | FIASA 2025 edition, p.47: petrol 9.03, diesel 11.73. FIASA 2024 edition, p.32: 8.76 and 11.81. JODI: 10.14 and 10.04 (lowest reliability code). Department: not published. Road Accident Fund levy: 24.4 bn litres for both fuels in the year to March 2025, up 0.7%, against 20.76 here.         | FIASA 2025 edition, flagged unverified               | Nigel: agree the 2024 sales source |
| <b>2024 diesel imports</b>           | Customs 10.793. Department trade report 10.8 (rounded). FIASA 14.793, a suspected misprint.                                                                                                                                                                                                       | Customs                                              | Nigel: review                      |
| <b>2018 and 2019 trade</b>           | 2018 diesel: customs 5.12, FIASA 6.11; petrol: 1.83 and 2.19. 2019 diesel: 5.59 and 5.86. 2018 exports are also lower in customs. FIASA and the department's 2018 balance share one earlier reading; today's customs data and UN Comtrade agree with each other. No missing month or tariff line. | Customs; likely a later customs revision, not proven | Nigel: review                      |
| <b>2022 diesel exports</b>           | Customs 0.89, FIASA 0.70.                                                                                                                                                                                                                                                                         | Customs                                              | Nigel: review                      |
| <b>2019 and 2020 energy balances</b> | Their import and export lines repeat the previous year's figures.                                                                                                                                                                                                                                 | Not used for trade                                   | Nigel: confirm                     |

In the other years from 2014, customs and FIASA differ by 0.2 bn litres or less; the full list is in the balance note. Before 2014 customs is in kilograms or mixed units, so FIASA is used.

# Production after 2021 and stock changes are missing, so the balance is not closed

Missing production and stocks | sales less net imports against production reported in the energy balance



A diesel gap of 3 to 4 bn litres a year remains a hypothesis to test. It is not added to demand. Needed to close: output by product and plant, a stock series, and matched periods.

# Diesel: rail diversion is capped near 13% by rail-friendly freight; yields move up

Diesel inputs | low / medium / high for 2030 and 2035 | 11 of 42 values have a proposed replacement

| Lever and unit                                      | Baseline                            | 2030<br>low / medium / high      | 2035<br>low / medium / high          | Rationale and source                                                                                                                                 |
|-----------------------------------------------------|-------------------------------------|----------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Road freight activity<br>2024 = 100                 | 100<br>980 Mt road payload,<br>2024 | 95 / 115 / 135                   | 95 / 135 / 150<br>was 95 / 135 / 175 | Stats SA tonnes: 2.4% a year over 2014-2024, at most 3.7% over eleven years. Only tonne-km figure found: 221 bn for 2013.                            |
| Road freight moved to rail<br>% of road freight     | 0<br>rail 161 Mt, 2024              | 0 / 3 / 9<br>was 0 / 10 / 20     | 0 / 7 / 13<br>was 0 / 20 / 35        | Rail Master Plan (2026): 250 Mt target is 9%, 280 Mt market appetite 12%. Freight Roadmap (2023): rail-friendly general freight left on road is 13%. |
| Diesel power generation<br>year to March 2024 = 100 | 100<br>5,143 GWh                    | 10 / 25 / 60                     | 5 / 15 / 50                          | Eskom: year to March 2026 is already 21. Agreed.                                                                                                     |
| New-vehicle efficiency gain<br>% a year             | 0.5 to 1.0<br>registered            | 0.5 / 0.75 / 1                   | 0.5 / 0.75 / 1                       | Equals the registered endpoints, which have no source. Agreed for now.                                                                               |
| Plant utilisation<br>% of capacity, all fuels       | 67<br>indicative, 2024              | 60 / 75 / 90                     | 60 / 75 / 90                         | Operators' reports: Secunda 53%, Natref 71%, Astron 84%. Secunda capacity corrected to FIASA's 150; output unchanged.                                |
| Diesel yield<br>% of refinery output                | 29<br>JODI 2024, low<br>reliability | 25 / 30 / 40<br>was 20 / 25 / 30 | 25 / 30 / 40<br>was 20 / 25 / 30     | JODI: 31% in 2023, 29% in 2024. Energy balances before the Durban closures: 40-42% of five fuels. Legacy 25%.                                        |
| Added refining capacity<br>thousand barrels a day   | 0                                   | 0 / 0 / 0                        | 0 / 0 / 400                          | Conditional illustration; no investment decision sourced.                                                                                            |

# Jet: movement highs exceed every year on record; fuel per movement can also rise

Jet inputs | low / medium / high for 2030 and 2035 | 6 of 36 values have a proposed replacement

| Lever and unit                                           | Baseline                           | 2030<br>low / medium / high            | 2035<br>low / medium / high            | Rationale and source                                                                                       |
|----------------------------------------------------------|------------------------------------|----------------------------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Aircraft movements</b><br>2024 = 100                  | 100<br>456,214 at ACSA<br>airports | 100 / 110 / 125<br>was 100 / 120 / 140 | 100 / 120 / 140<br>was 100 / 145 / 190 | ACSA: 2019 was 112 and the 2016 peak 125. Proposed highs of 140 and 190 exceed every year on record.       |
| <b>Jet fuel per movement</b><br>2024 = 100               | 100<br>4,090 litres                | 80 / 90 / 113<br>was 80 / 90 / 100     | 65 / 80 / 113<br>was 65 / 80 / 100     | Sales over movements: the 2013-2019 average was 4,608 litres (113). High case should allow a return to it. |
| <b>Sustainable fuel share</b><br>% of jet volume         | 0<br>unsourced                     | 0 / 2 / 5                              | 0 / 5 / 10                             | No South African blend requirement sourced. Agreed as a sensitivity.                                       |
| <b>Plant utilisation</b><br>% of capacity, all fuels     | 67<br>indicative, 2024             | 60 / 75 / 90                           | 60 / 75 / 90                           | Same setting as diesel and petrol; count once.                                                             |
| <b>Jet yield</b><br>% of capable throughput              | 10<br>legacy                       | 5 / 10 / 15                            | 5 / 10 / 15                            | Jet was 5-8% of the five fuels produced, 2017-2021 energy balances. Agreed.                                |
| <b>Added refining capacity</b><br>thousand barrels a day | 0                                  | 0 / 0 / 0                              | 0 / 0 / 400                            | Jet capability of any restart is not established.                                                          |

# Petrol: mileage and growth sit above the evidence; electric cases now follow named markets

Petrol inputs | low / medium / high for 2030 and 2035 | 22 of 42 values have a proposed replacement

| Lever and unit                                            | Baseline                         | 2030<br>low / medium / high                              | 2035<br>low / medium / high                              | Rationale and source                                                                                                             |
|-----------------------------------------------------------|----------------------------------|----------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Passenger mileage<br/>km a vehicle a year</b>          | 17,000<br>placeholder            | 12,900 / 14,500 / 17,000<br>was 13,600 / 17,000 / 20,400 | 12,900 / 14,500 / 17,000<br>was 13,600 / 17,000 / 20,400 | Stone et al. (2018): 14,457 km for petrol cars. 2023 sales over registered petrol vehicles imply about 12,900.                   |
| <b>Real GDP growth<br/>% a year</b>                       | 0.5<br>2024 actual               | 0.7 / 1.6 / 2<br>was 1 / 2 / 3                           | 0.7 / 1.6 / 2<br>was 1 / 2 / 3                           | Stats SA: 0.7% a year over 2014-2024, 1.6% over 2010-2019. Treasury: 2.0% by 2028. Registered cases 1.0, 1.6.                    |
| <b>Battery electric share<br/>% of new sales</b>          | 0.2<br>1,088 sold, 2025          | 0.5 / 4 / 20<br>was 5 / 15 / 30                          | 2 / 15 / 40<br>was 15 / 35 / 60                          | IEA Global EV Outlook 2026, 2025 shares: India nearly 4%, Indonesia 15%, Turkiye over 20% (from 1% in 2022), Vietnam nearly 40%. |
| <b>New-vehicle efficiency gain<br/>% a year</b>           | 1.0 to 1.5<br>registered         | 1 / 1.25 / 1.5                                           | 1 / 1.25 / 1.5                                           | Equals the registered endpoints, which have no source. Agreed for now.                                                           |
| <b>Plant utilisation<br/>% of capacity, all fuels</b>     | 67<br>indicative, 2024           | 60 / 75 / 90                                             | 60 / 75 / 90                                             | Same setting as diesel and jet; count once.                                                                                      |
| <b>Petrol yield<br/>% of refinery output</b>              | 54<br>JODI 2024, low reliability | 45 / 50 / 55<br>was 35 / 45 / 55                         | 45 / 50 / 55<br>was 35 / 45 / 55                         | JODI: 51% in 2023, 54% in 2024. Energy balances 2017-2021: 47-50% of five fuels. Legacy 45%. Set with diesel.                    |
| <b>Added refining capacity<br/>thousand barrels a day</b> | 0                                | 0 / 0 / 0                                                | 0 / 0 / 400                                              | Conditional illustration; no investment decision sourced.                                                                        |

# Nine levers added: price, backup generation, freight electrification, hybrids, rail and neighbours

Added by the analyst; not in Nigel's file | low / medium / high for 2030 and 2035 | proposals for review

| Lever and unit                                                            | Baseline                 | 2030<br>low / medium / high | 2035<br>low / medium / high | Rationale and source                                                                                                                           |
|---------------------------------------------------------------------------|--------------------------|-----------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Real diesel price<br/>2024 = 100</b>                                   | 100<br>wholesale, inland | 80 / 100 / 140              | 80 / 100 / 140              | Department and CEF prices over CPI: 76-120 in 2011-2025, 143 in October 2026. Boshoff (2012): response -0.13.                                  |
| <b>Private backup generation<br/>bn litres a year</b>                     | Not measured             | 0 / 0 / 0.5                 | 0 / 0 / 0.5                 | Diesel sales fell 1.2 bn litres in 2024 as load-shedding ended; grid turbines explain about 0.7. Indicative.                                   |
| <b>Electric share of new trucks<br/>% of new truck sales</b>              | Not reported             | 0 / 0.4 / 3                 | 0 / 3 / 9                   | No South African data. 2025 benchmarks: Brazil 0.4%, Europe 3%, world 9%, China 25% (IEA; ICCT). High case is about 2.5% of the fleet by 2035. |
| <b>Electric share of new light commercial<br/>% of new sales</b>          | Not reported             | 0 / 1 / 5                   | 0 / 5 / 10                  | 2025 benchmarks: India 1%, Europe 10%, China 14% (IEA). Bakkies differ from vans, so loose comparators.                                        |
| <b>Real petrol price<br/>2024 = 100</b>                                   | 100<br>retail, inland    | 80 / 100 / 120              | 80 / 100 / 120              | 79-109 in 2011-2025, 121 in October 2026. Boshoff (2012): response -0.5, so the high case is about -10%.                                       |
| <b>Plug-in hybrid share<br/>% of new sales</b>                            | 0.5<br>2,810 sold, 2025  | 0.5 / 1.2 / 2.1             | 0.5 / 2 / 3.7               | naamsa: 0.02% in 2022 to 0.47% in 2025. Extrapolates South Africa's own sales; not a forecast.                                                 |
| <b>Conventional hybrid share<br/>% of new sales</b>                       | 2.1<br>12,818 sold, 2025 | 2.1 / 4.4 / 6.7             | 2.1 / 6.7 / 11.3            | naamsa: 0.77% in 2022 to 2.15% in 2025. Extrapolates South Africa's own sales; not a forecast.                                                 |
| <b>Rail passenger journeys<br/>million a year</b>                         | 74<br>2024               | 100 / 180 / 320             | 100 / 250 / 320             | Stats SA: 317 in 2017, 175 in 2019, 19 in 2022, 103 in 2025. Effect in litres cannot yet be stated.                                            |
| <b>Exports to nine neighbours<br/>bn litres a year, petrol and diesel</b> | 1.48<br>2024             | 0.7 / 1.4 / 1.7             | 0.7 / 1.4 / 1.7             | SARS: 2.66 in 2019, 1.37 in 2025; Botswana 1.05 to 0.68. Terminal throughput, not South African demand.                                        |

Source: fuel\_lever\_response\_2026-10-07.md on manish-branch; Nigel's proposed inputs of 7 October are not edited. Replacements are the analyst's proposals, pending review.

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## Sourced starting values lower 2024 diesel demand by 0.63 bn litres

Agriculture and industry | diesel, billion litres a year | proposed as provisional starting values

| Sector      | Earlier placeholder | On the branch                                                   | Range in the evidence                                                                        | Proposed                           | Effect in 2024; 2035             |
|-------------|---------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------|----------------------------------|
| Agriculture | 0.70<br>no source   | 1.06 for 2021<br>energy balance, agriculture and forestry       | 0.90-1.09 in eight of ten years, 2012-2021;<br>1.89 and 1.88 in 2016 and 2017                | 1.06<br>provisional, about +/- 0.1 | +0.36; +0.36                     |
| Industry    | 2.50<br>no source   | 1.50 for 2021<br>energy balance, industry; mining is 1.29 of it | 1.32-1.93 over 2012-2021. Mining volume is 7% lower in 2024 than 2021, which would give 1.40 | 1.50<br>provisional, about +/- 0.3 | -0.99; -1.06 (high), -1.00 (low) |
| Both        | 3.20                | 2.56                                                            |                                                                                              | 2.56                               | -0.63; -0.69 (high), -0.64 (low) |

Why 2021 and not an average: it is the latest balance and one consistent year for both sectors. A 2018-2021 average gives 0.98 and 1.68; the industry average is pulled up by the years before mining output fell.

Decision for Nigel: accept both as provisional, or take the activity-adjusted 1.40 for industry. Both keep the needs-verification flag.

# The totals show no double counting from the two sectors; the excess is in power generation

## Overlaps and double counting | model diesel for 2024 by segment against the evidence, billion litres

| Model segment    | Model 2024 | Evidence                                                                                          | Overlap or gap                                                                                                                         |
|------------------|------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Road vehicles    | 8.50       | Balance 2021: road 6.33; commercial and public services 4.30; together 10.63                      | The two balance lines swap volume between years while their sum holds, so the split shows who bought the fuel, not where it was burned |
| Agriculture      | 1.06       | Balance 2021: 1.06                                                                                | Includes farm vehicles on public roads, which are also in the registered fleet. Size unknown                                           |
| Industry         | 1.51       | Balance 2021: 1.50; mining 1.29, construction 0.05                                                | Mine haul trucks are unregistered, so no overlap; mines' road vehicles do overlap                                                      |
| Power generation | 3.58       | Eskom: 0.94 (year to March 2023), 0.68 (to March 2025); about 1.6 for all plants in the peak year | Not an overlap: the model is at least 2 bn litres above reported burn                                                                  |
| Marine gas oil   | 0.77       | No observed figure                                                                                | Not established whether ships' diesel is inside inland sales                                                                           |
| Total            | 15.42      | Recorded sales: 12.91 (2023); 11.73 (2024, unverified)                                            | Model is 2.5 to 3.7 above recorded sales                                                                                               |

Road vehicles, agriculture and industry are 11.07 in the model against 13.41 of final consumption in the 2021 balance, so those three are short, not double counted. At most 2.56 could overlap with road vehicles; no source sizes it.

## Lesedi: eight loading bays cap the high case near 2.4 turns a month, not 3

Vopak Lesedi, Jameson Park | proposed site assumptions | estimates are to confirm with the client

| Item                           | Value                                                                                                         | Basis                                                 | Source                                                            |
|--------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------|
| <b>Design capacity</b>         | 140,000 m3: petrol 40,000, diesel 80,000, one 20,000 tank for either. New tanks in service since October 2025 | Evidence                                              | NERSA licence amendment, 6 August 2025; Vopak Q3 2025 release     |
| <b>Operational capacity</b>    | 124,000 m3: 88,000 in the six original tanks; 18,000 assumed for each new tank                                | Evidence; new tanks estimated, to confirm with client | Vopak licence application to NERSA, 2020, p.8-9                   |
| <b>Receipts</b>                | Pipeline only, at most 1,500 m3 an hour                                                                       | Evidence                                              | Licence application, p.9                                          |
| <b>Dispatch</b>                | At most 1,500 m3 an hour; 80% road, 20% pipeline. 8 bays at 125 m3 an hour                                    | Evidence                                              | Licence application, p.9; NERSA 2025                              |
| <b>Practical road dispatch</b> | About 300,000 m3 a month                                                                                      | Estimate, to confirm with client                      | Bays pumping half the time, 20 hours a day                        |
| <b>Monthly turns</b>           | 1 / 2 / 2.4 (low / base / high), on operational capacity                                                      | Estimate, to confirm with client                      | Base from Vopak's Durban target; high from road dispatch          |
| <b>Annual handling</b>         | 1.5 / 3.0 / 3.6 million m3                                                                                    | Calculated                                            | Pack today: 1.7 / 3.4 / 5.0 on gross capacity and 1 / 2 / 3 turns |

Vopak's website publishes none of the operating data. Still needed from Vopak: twelve months of outbound volume by product and mode, the new tanks' operational capacity and loading hours.

# Durban: fuel tanks are under half the site, so handling is far below the gross-capacity figure

Vopak Durban, Island View | proposed site assumptions | estimates are to confirm with the client

| Item                        | Value                                                                       | Basis                                        | Source                                                |
|-----------------------------|-----------------------------------------------------------------------------|----------------------------------------------|-------------------------------------------------------|
| Gross capacity              | 360,246 m3 for petroleum, chemicals and oleochemicals                       | Evidence                                     | Vopak terminal page, checked 6 October 2026           |
| Built for petrol and diesel | 162,000 m3 in ten tanks; a floor, older tanks not counted                   | Evidence of a floor, to confirm with client  | Vopak news release, 17 February 2017                  |
| Operational capacity, fuel  | 146,000 m3                                                                  | Estimate, to confirm with client             | 90% of design, as at Lesedi                           |
| Receipts and dispatch       | Vessel (five berths), pipeline, rail, truck. No rate published for any mode | Evidence; rates not found                    | Vopak terminal page                                   |
| Pipeline to Gauteng         | 148,000 m3 a week for all shippers and products                             | Evidence                                     | Transnet Pipelines, 24 inch trunk line                |
| Monthly turns               | 1 / 2 / 3 (low / base / high), on operational fuel capacity                 | Target, not realised; to confirm with client | Vopak to NERSA: 'targeting 2-3 throughputs in Durban' |
| Annual handling, fuel       | 1.8 / 3.5 / 5.3 million m3                                                  | Calculated                                   | Pack today: 4.3 / 8.6 / 13.0 on gross capacity        |

The high case needs 438,000 m3 a month, 68% of the whole trunk line if it all went inland by pipe. Durban-to-Lesedi volume is handled at both sites and must be counted once.

# All five priorities have outputs to review

Status against the five priorities of 7 October | ready, proposed or partial

| Priority           | Status                  | Ready to review                                                                                                                                                      | Not done or not available                                                      |
|--------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 1 National balance | Ready, with stated gaps | Balance rebuilt on customs trade with tests; source differences listed; 2018-2019 traced as far as data allows                                                       | Production by product after 2021 and stocks exist in no source found           |
| 2 Forecast levers  | Proposed                | 20 levers with baseline, rationale and source; 39 of 120 values with a replacement; nine levers added                                                                | Some baselines are placeholders; freight is in tonnes                          |
| 3 Sector baselines | Proposed                | Agriculture 1.06 and industry 1.50 bn litres, with the forecast effect and an overlap table                                                                          | On-road overlap not sized; power generation is 2 bn litres above reported burn |
| 4 Tank handling    | Proposed                | Lesedi and Durban site assumptions; eight values marked to confirm with the client                                                                                   | No Vopak operating data; other operators' sites not done                       |
| 5 Hand-back        | Ready                   | This deck, the hand-back note, a 21-item decision log, the workshop workbook with history from 2012, and 288 figures in the current pack checked with none differing | Vehicle stock by age and by fuel within class is in no source held             |

**New evidence today: Road Accident Fund accounts give 24.4 bn litres of petrol and diesel levied in the year to March 2025, against 20.8 recorded as sold, and show volumes rising where FIASA shows a fall.**

# Decision log: 3 ready, 9 proposed, 9 unresolved

## Decisions for Nigel | the nine that are unresolved, with the analyst's proposal | full log in the csv

| Decision                                                                  | Evidence in brief                                                                                                                                                                                    | Proposal                                               |
|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>D01 Which 2024 national sales figure to use</b>                        | Department has published nothing after 2023. FIASA 2025 edition: petrol 9.03, diesel 11.73 bn litres; FIASA 2024 edition: 8.76 and 11.81. Road Accident Fund levy shows volumes up 0.7%, not down 5% | Start the forecast from 2023; carry 2024 as unverified |
| <b>D05 Treatment of the 2014 and 2018 provincial differences</b>          | Provincial sum is above the national file by 0.13-0.14 bn litres in 2014; 2018 quarter 1 district sheet is missing districts                                                                         | National file stands; both records kept                |
| <b>D08 Secunda utilisation</b>                                            | Capacity corrected to 150 thousand barrels a day with utilisation re-based so output is unchanged; model gives 62 for 2024, Sasol reports 80                                                         | Raise utilisation to match reported output             |
| <b>D09 Reference case</b>                                                 | No medium case is agreed; medium values are proposals                                                                                                                                                | Agree after the ranges                                 |
| <b>D11 Power generation starting value</b>                                | Model has 3.58 bn litres for 2024; Eskom reports 0.68-0.94 and all plants about 1.6 in the peak year                                                                                                 | Recalibrate to reported burn at 0.31 litres per kWh    |
| <b>D12 Allowance for on-road use inside agriculture and mining diesel</b> | No published source sizes it; 20% would be about 0.47 bn litres; Road Accident Fund rebates cover 1.46 bn litres across all rebated sectors                                                          | None applied; ask SARS for refund data by sector       |
| <b>D13 Whether ships' diesel sits inside the inland diesel total</b>      | Model carries 0.77 bn litres of marine gas oil; no observed figure                                                                                                                                   | Decide scope before reconciling to sales               |
| <b>D17 Send the data request to Vopak</b>                                 | Eight values marked to confirm with client; Vopak's website publishes no operating data                                                                                                              | Send now                                               |
| <b>D21 Marine baseline elasticities and road parameters</b>               | Carried over from 6 October; no new evidence                                                                                                                                                         | Unchanged                                              |

# The workshop workbook now carries sourced history from 2012 and labelled provincial estimates

Demand\_baseline\_workshop\_2026\_10\_08\_history.xlsx | built by one command | Nigel's sheets unchanged

| Sheet added               | What it holds                                                                                                                                                             | What it shows                                                                                                            |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>History</b>            | Department sales by province, customs imports and exports, refinery production, the balance and the Road Accident Fund count, 2012-2025; totals and balances are formulas | Net imports of petrol and diesel rose from 3.7 bn litres (2014) to 15.7 bn (2022). Production by product stops at 2021   |
| <b>History, section 6</b> | Estimated provincial shares and volumes for 2023-2025, kept apart from observed rows                                                                                      | Quarter 1 2023 shares are the best of six methods tested: 2.5 share points misallocated for petrol, 5.3 for diesel       |
| <b>Sector history</b>     | Mining, manufacturing and agriculture diesel beside their activity measures, with litres per unit of activity                                                             | Mining swings between 11 and 17 million litres per index point; indicative 2024: mining 1.40, agriculture 0.96 bn litres |
| <b>Vehicle history</b>    | Registered vehicles, new sales, retirements, drivetrain and fuel split beside the model's settings                                                                        | Cars are 65-71% of new sales against 78% in the model; petrol per vehicle is 1,056 litres against 1,615 implied          |
| <b>Checks</b>             | 66 values on the Evidence sheet recomputed from registered inputs                                                                                                         | All match                                                                                                                |
| <b>History sources</b>    | Publisher, link, original file, extract and refresh command for each dataset                                                                                              |                                                                                                                          |

One change to an existing sheet: source selection for 2022 and 2023 is set to the department instead of FIASA. 2024 stays on FIASA, unverified. 2025 has no national sales figure from any source.

# Review the branch and decide

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manish-branch | checks pass | nothing  
merged to main

7 October 2026

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